**Physics applications of definite integrals**

Here is a quick reference for the answers to this problem set.

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1. The kinetic energy of the second hand is $5.483 \times 10^{-7}\,\mathrm{J}$.
2. The total mass of the oil slick is $3.1386 \times 10^{6}\,\mathrm{kg}$.
3. The energy to launch the CubeSat to the moon is $1.227 \times 10^{8}\,\mathrm{J}$.
4. The energy to push the movable magnet is $k \cdot 13.33\,\mathrm{J}$.
5. The total mass of the atmosphere is $5.162 \times 10^{18}\,\mathrm{kg}$.
6. The gravitational attraction between the rod and the ball is $\dfrac{GMm}{a(a+\ell)}\mathrm{N}$.